

ORIGINAL ARTICLE

Open Access



A method for siting adsorption-based direct air carbon capture and storage plants for maximum CO₂ removal

D. Rezo^{1*} , P. Postweiler¹, M. Engelpracht¹, L. Meuleneers¹ and N. von der Aßen¹

Abstract

Adsorption-based direct air carbon capture and storage (DACCS) is an emerging approach to mitigate climate change by removing CO₂ from the atmosphere. Recent studies show separately that thermodynamic and environmental performance strongly depend on regional ambient conditions and energy supply but neglect regional CO₂ storage potentials. To assess DACCS performance holistically, a detailed global analysis is needed that accounts for the interplay of regional ambient conditions, energy supply, and CO₂ storage potential. Hence, we present a novel method for the optimal siting of DACCS plants derived from optimising a dynamic process model that uses global hourly weather data and regionalised data on electricity supply and CO₂ storage potential. The carbon removal rate (CRR) measures the climate benefit and describes the speed at which a DACCS plant generates net negative emissions. First, we assume that CO₂ storage is possible everywhere. For four electricity supply scenarios, we show that the optimal siting of DACCS significantly increases the CRR when comparing the best and worst locations in each scenario: For a DACCS plant with a nameplate capture capacity of 4 kt CO₂ y⁻¹, the CRR can be increased by 63% from 2.16 to 3.53 kt CO₂ y⁻¹ when using photovoltaic, and by 39% from 2.95 to 4.1 kt CO₂ y⁻¹ when using wind power. Assuming a carbon-free electricity supply, the CRR varies between 3.17 and 4.17 kt CO₂ y⁻¹ (32%). Second, we significantly narrow down optimal locations for DACCS considering regional CO₂ storage potential through CO₂ mineralisation. Overall, accounting for the interplay of regional DAC performance, energy supply, and CO₂ storage potential can significantly improve DACCS siting.

Keywords Direct air carbon capture and storage, Assessment framework, Life cycle GHG emissions, Geospatial assessment, Dynamic process model, Process optimization

1 Introduction

The major driver of climate change is the increasing concentration of greenhouse gases (GHGs) in the atmosphere [1]. Hence, reducing GHG emissions is crucial, particularly using renewable energy sources such as wind and solar power. Still, hard-to-abate emissions will remain in sectors like aviation and agriculture [2]. Active

removal of atmospheric CO₂ – so-called negative emissions – is necessary to offset these hard-to-abate emissions [2, 3].

Direct air carbon capture and storage (DACCS) is a promising approach for creating negative emissions by removing CO₂ from the atmosphere and permanently storing it in geological formations. The central process of DACCS is direct air capture (DAC). Most DAC approaches discussed in the literature use adsorption- or absorption-based processes to capture CO₂ from the atmosphere [4]. Typically, adsorption-based processes require lower regeneration temperatures (80–130 °C) than absorption-based processes (up to 900 °C). The

*Correspondence:

D. Rezo
daniel.rezo@ltt.rwth-aachen.de

¹ Institute of Technical Thermodynamics, RWTH Aachen University, 52062 Aachen, Germany

energy demand highly depends on the exact DAC process: According to a recent literature review, heat demand tends to be lower for absorption-based processes compared to adsorption-based processes (5.25–8.8 vs. 4–11.8 GJ (t CO₂)⁻¹), whereas electricity demand tends to be higher for absorption-based processes (0.82–1.52 vs. 0.4–1.4 GJ (t CO₂)⁻¹) [5]. Still, the lower temperatures of adsorption-based processes enable the integration of renewable energy via heat pumps [4, 6]. In the temperature range of 80–130 °C, air heat pumps can achieve a coefficient of performance of 2 to 3 [7], thus significantly lowering the overall energy consumption of an adsorption-based DAC plant.

1.1 Numerical models of adsorption-based DAC

Due to the low concentration of CO₂ in the atmosphere (approximately 420 ppm) adsorption-based DAC plants are designed to process enormous amounts of air. Therefore, air conditions are a major input variable. In particular, air moisture influences CO₂ adsorption on solid surfaces [8–10]. Numeric models have been developed to address the competitive adsorption of CO₂ and other species, e.g. water, and changes in the reaction pathways of CO₂ on the surface of the sorbent [8, 10–13]. Recent numerical studies of DAC have examined the impact of ambient conditions, such as temperature and humidity, on thermodynamic performance metrics, e.g., specific energy demand (SED) and productivity [10, 13–17]. Here, productivity indicates the gross mass of CO₂ captured per unit of time. Stampi-Bombelli et al. [13] and Young et al. [10] investigated the amine-functionalised sorbents APDES-NFC and Lewatit, respectively. The authors provided models for the adsorption equilibrium and adsorption kinetics of CO₂ and H₂O at varying ambient temperatures and relative humidities. For both materials, increased relative humidity increases CO₂ capacity. Using a dynamic DAC model, Young et al. showed that the increase in CO₂ capacity can reduce SED and increase productivity. However, at very high ambient humidity levels the energy duty for H₂O desorption can outweigh positive effects of H₂O co-adsorption.

1.2 Impact of ambient conditions on adsorption-based DAC

By optimisation of a dynamic model, Wiegner et al. [17] and Schellevis [15] determined the optimal SED and productivity of DAC for sets of ambient temperature and humidity. Wiegner et al. utilised the optimisation results to create a performance diagram, i.e., a linearised model representation. The performance diagram allows for a simplified assessment of DAC at varying ambient conditions.

Sendi et al. [16] conducted a geospatial analysis of a DAC plant by integrating global hourly weather data into a dynamic DAC model using the amine-functionalised sorbent Lewatit. The DAC plant was powered by electricity, with heat supplied by a heat pump. The authors presented world maps illustrating regional SED and productivity of the investigated DAC plant. Their analysis showed that productivity is lowest in hot and humid regions and highest in cold and dry regions, with the mean productivity increasing by 10% between the regions. On the contrary, SED is highest in cold and humid regions (2.15 MWh_{el} (t CO₂)⁻¹) and lowest in hot and dry regions (1.64 MWh_{el} (t CO₂)⁻¹), representing a decrease of 24%. The decrease can be partly attributed to the decreasing efficiency of the heat pump with lower ambient temperatures. Overall, selecting optimal locations for DAC must balance SED and productivity.

1.3 Life cycle assessments of adsorption-based DACCS

Recent life cycle assessments (LCA) have investigated the environmental performance of DACCS [18–21]. For this purpose, many LCA studies use the carbon removal efficiency (CRE) as the key metric. The CRE represents the captured mass of CO₂ after subtracting the CO₂ emissions from the DACCS process relative to the captured mass of CO₂. The emissions from the DACCS process include emissions from construction, operation, energy supply, CO₂ transport and storage, and end-of-life of the entire DACCS plant. It was shown that an integrated system of DACCS and a heat pump could achieve CRE values higher than 97% by using wind electricity [18]. In contrast, using fossil electricity as driving energy may result in negative CRE values, i.e., additional instead of negative emissions [18, 20, 21]. Therefore, the carbon footprint of the energy supply significantly affects the CRE and, consequently, the environmental performance of DACCS.

Terlouw et al. [21] investigated the CRE of DACCS based on a static DAC model at eight locations spread across different countries using five energy system layouts. The authors found that DACCS could achieve CRE values of 79 – 97% in all investigated locations when using low-carbon energy sources. Moreover, the authors found that CO₂ transport and storage significantly contribute to GHG emissions of a DACCS plant. Hence, CO₂ storage should be located close to the DAC plant. For an integrated system of DACCS and a heat pump, the authors conducted a global analysis using country-specific data for the carbon footprint of grid electricity. The global analysis showed that the CRE varied widely from 9% in Greece to 95% in Norway. However, the static DAC model neither accounted for the effect of regional climate

on DAC performance nor effects of intrinsic dynamic behaviour of DAC.

1.4 Environmental process optimisation of adsorption-based DACCS

Recently, we integrated LCA data into a dynamic DACCS model using the amine-functionalised cellulose APDES-NFC as sorbent [22]. In this way, our model enables the optimisation of DACCS regarding combined thermodynamic and environmental objective functions. We demonstrated a trade-off between CRE and productivity through multi-objective optimisation. For example, when productivity increases from 3 to 4 kt CO₂ y⁻¹, pareto-optimal CRE decreases from 65.5 to 57.3%.

1.5 Conclusions and contribution

In summary, recent assessments of adsorption-based DAC showed that regional ambient conditions greatly affect thermodynamic performance, such as productivity and SED. Recent LCA studies indicated that the environmental performance of adsorption-based DACCS strongly depends on the carbon footprint of the energy supply, which varies strongly across global regions. Our recent results demonstrated that the optimal thermodynamic and environmental performance of adsorption-based DACCS are interdependent. Furthermore, CO₂ storage should be located close to the DAC plant to achieve negative emissions with DACCS. Hence, a global analysis of adsorption-based DACCS is needed that accounts for the impact of ambient conditions and regional energy supply on the thermodynamic and environmental performance of DACCS. Additionally, the regional CO₂ storage potential must be considered, since transport can significantly contribute to the GHG emissions of a DACCS plant.

Here, we present a novel method for an integrated global analysis of DACCS, energy supply, and CO₂ storage potential (Sect. 2). The ultimate purpose of our method is to identify sites for DACCS that maximise its climate benefit, measured by the carbon removal rate (CRR). The CRR quantifies the net mass of CO₂ that can be removed per unit of time by a DACCS plant with a given size and design. As a case study, we investigate a heat-pump-driven adsorption-based DAC system integrated with CO₂ storage. In the first step, we assume that CO₂ storage is possible near every location worldwide. Thus, we focus our analysis on the impact of ambient conditions and energy supply on DAC, the main process of DACCS. In the second step, we account for regional CO₂ storage potential further narrowing down potential DACCS sites. To address the trade-off between CRE and productivity, we use the CRR as a performance metric. By optimisation of a dynamic model, we calculate the

maximal CRR for varying ambient conditions and different carbon footprints of the electricity supply (Sect. 3). Subsequently, we calculate regionalised carbon footprints for four electricity supply scenarios (Sect. 4). We use global hourly weather data and the regionalised data on carbon footprints and CO₂ storage potential to assess the CRR globally, identifying optimal sites for DACCS operation (Sect. 5). Thus, our method can be regarded as the basis for identifying optimal sites for adsorption-based DACCS plants worldwide to maximise net CO₂ removal.

2 Global assessment of the carbon removal rate of DACCS plants

To assess the CRR of adsorption-based DACCS globally, we use the four-step approach illustrated in Fig. 1. In the following, adsorption-based DAC(CS) is referred to as DAC(CS), as our analysis is limited to an adsorption-based process.

In the first step, we derive a performance diagram of the CRR. For this purpose, we optimise a dynamic DACCS model integrated with a heat pump for different sets of temperatures, relative humidities, and electricity carbon footprints. The performance diagram is a linearised representation of optimal process designs based on the dynamic model, allowing for a quick calculation of maximal achievable CRR-values for different ambient temperatures, relative humidity levels, and electricity carbon footprints.

In the second step, we determine regionalised carbon footprints for four exemplary electricity supply scenarios: (a) An ideal electricity supply with zero emissions, (b) grid electricity, (c) electricity from photovoltaics (PV), and (d) electricity from wind turbines. Scenario (a) isolates the impact of ambient conditions on the CRR. Hence, scenario (a) allows for assessing the climate impact on CRR separately from the electricity supply. Scenario (b) uses grid electricity to assess the viability of DACCS when powered by the electricity grid in different countries. As DACCS should ideally be powered by renewable energy sources, scenarios (c) and (d) assess DACCS powered by solar PV and wind electricity.

For scenario (a), we determine country-specific carbon footprints of grid electricity based on data from the LCA database ecoinvent [23]. Ecoinvent provides data on the carbon footprint of grid electricity in 136 countries from 2014 to 2022. For scenarios (c) and (d), we calculate the regionalised carbon footprints of PV and wind electricity. For this purpose, we determine the average regional energy yield of a PV and a wind power plant. Subsequently, we calculate the regionalised carbon footprints of PV and wind electricity based on LCA data from ecoinvent. More information is given in the supporting information.

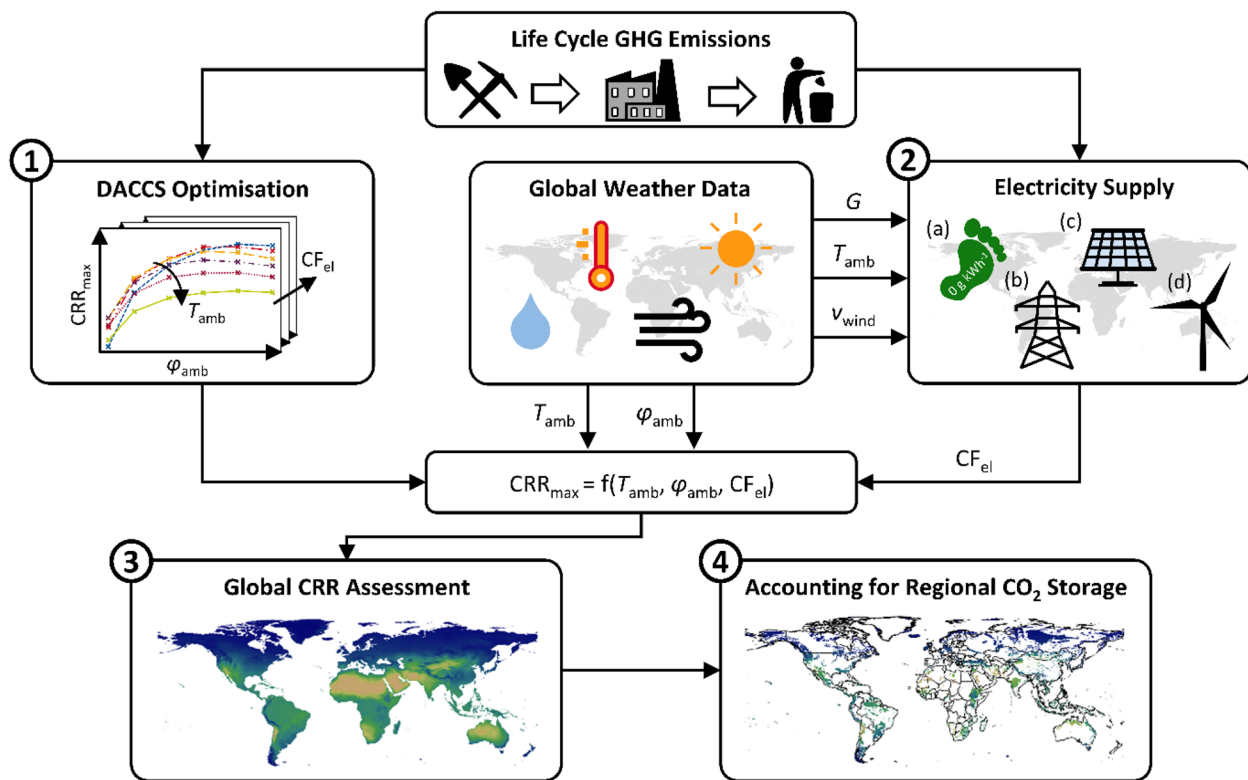


Fig. 1 Four-step approach for the geospatial assessment of the carbon removal rate (CRR) of adsorption-based direct air carbon capture and storage (DACCS): (1) A performance diagram of the CRR is generated by optimisation of a dynamic DACCS model. (2) Regional carbon footprints of grid, PV, and wind electricity (CF_{el}) are determined based on global weather data and data on life cycle GHG emissions. (3) The performance diagram, regional weather data, and electricity carbon footprints are combined for a global assessment of the CRR. (4) Regional data on CO₂ storage potential via CO₂ mineralisation is used to further narrow down optimal sites for DACCS

In the third step, we use hourly weather data from 2015 to 2019 and the regional carbon footprints of the four electricity supply scenarios to determine the CRR of DACCS on a global level. In the fourth step, we narrow down potential sites for DACCS by considering regional CO₂ storage potential via CO₂ mineralisation as exemplary storage method. For this purpose, we intersect the regional DACCS performance of the four investigated electricity supply scenarios with a global map on CO₂ storage potential via CO₂ mineralisation published by the company Carbfix [24, 25].

3 Modelling and optimisation of a DACCS plant

3.1 Modelling of a DACCS plant

We investigate a DACCS plant using a dynamic model implemented and validated based on literature data by our research group [22] in the programming language Modelica using our open-source Modelica library SorpLib [26], freely available on GitLab [27]. The model describes the inherent process dynamics of adsorption-based DACCS caused by, e.g. the rapid temperature and pressure changes in the process. Additionally, the model

includes life cycle assessment data accounting for the life cycle GHG emissions of plant construction, end of life, and electricity and sorbent supply. Thus, our model enables the optimisation of DACCS regarding combined thermodynamic and environmental objective functions, such as the carbon removal rate. As the DACCS model remains unchanged for the present study, we do not provide a detailed model description and only address key aspects in the following. For a detailed model description please refer to our previous work [22]. The DACCS plant comprises three key components (Fig. 2): (1) DAC unit, (2) heat pump, (3) compressor and CO₂ injection. All components are powered by electricity ($\dot{W}_{el}$).

The central component of the DACCS plant is the DAC unit, which is modelled as a single adsorption column using APDES-NFC, an amine-functionalised cellulose, as sorbent. Our model is flexible, so that other sorbents could be modelled when data on adsorption equilibria and kinetics is available. We use APDES-NFC as it is investigated as a promising candidate for DAC in the current literature [13, 14]. Moreover, it is described to be similar to the commercial sorbent used by Climeworks,

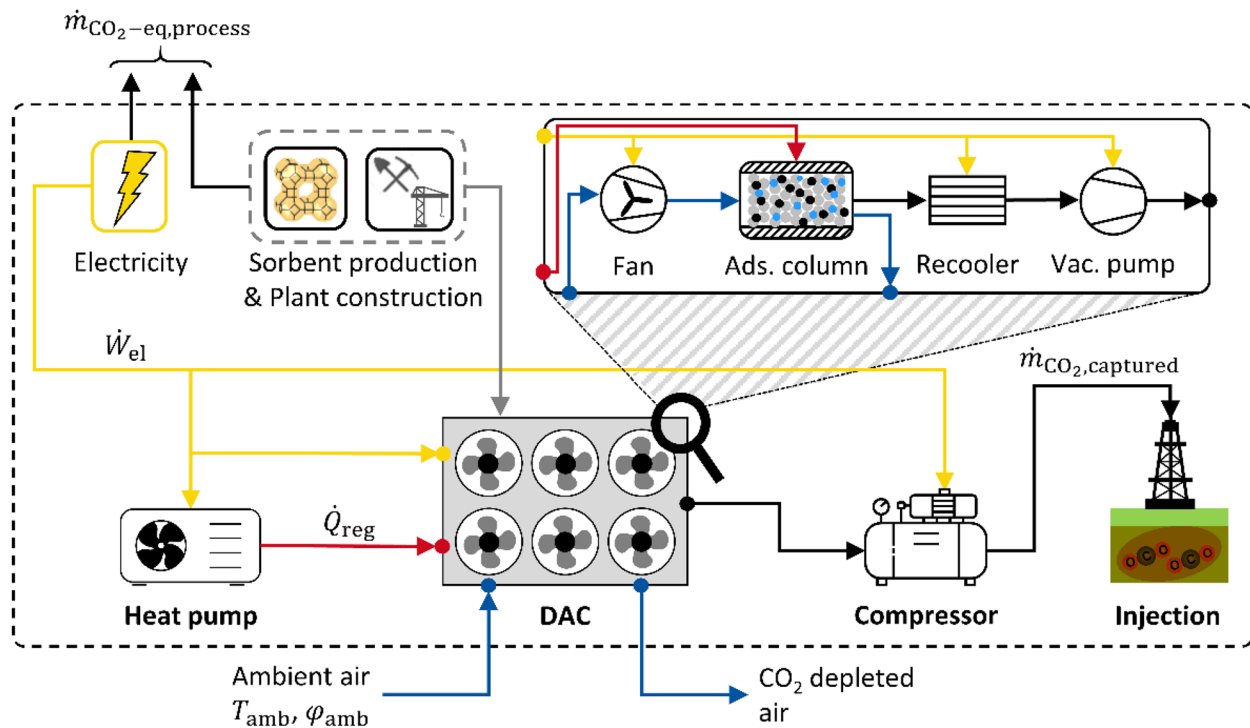


Fig. 2 Model of the investigated adsorption-based direct air carbon capture and storage (DACCS) plant powered by electricity $\dot{W}_{el}$. Ambient air enters the direct air capture (DAC) unit, which comprises of a fan, an adsorption column, a recooler, and a vacuum pump at ambient temperature (T_{amb}) and humidity (φ_{amb}). For regeneration, the heat flow rate $\dot{Q}_{reg}$ is supplied to the adsorption column and the adsorption column is evacuated via the electrically driven vacuum pump. The process emissions $\dot{m}_{CO_2-eq,process}$ include emissions from construction and end-of-life of the DACCS plant as well as sorbent production and life cycle GHG emissions from the electricity supply. The gross mass of CO_2 captured by the DACCS plant is represented by $\dot{m}_{CO_2,captured}$. A detailed model description can be found in our recent study [22]

the largest company applying adsorption-based DACCS to date [28]. Data on adsorption equilibria and kinetics is based on work by Stampi-Bombelli et al. [13]. The single-column model was validated based on literature data from small-scale experiments and linearly scaled to achieve a maximal productivity of $4 \text{ kt } CO_2 \text{ y}^{-1}$ at a reference ambient temperature of 20°C and relative humidity of 49% [22]. The productivity matches the Orca plant from Climeworks. The Orca plant is also the basis for the LCA data integrated into the dynamic model [18]. For a detailed description of the model validation and scale-up, please refer to our previous study [22].

In the present study, we employ a DAC process based on a temperature-vacuum swing adsorption cycle, as this cycle is also used at the Orca plant. The cycle has five distinct phases: (1) an adsorption phase, (2) a blow down phase, (3) a heating phase, (4) a desorption phase, and (5) a cooling phase.

In the adsorption phase, a fan pushes air at ambient temperature (T_{amb}) and relative humidity (φ_{amb}) through the adsorption column. The sorbent adsorbs CO_2 and H_2O from the airflow. In the subsequent blow down phase, a vacuum pump evacuates the adsorption column to a

desorption pressure below 5000 Pa. Upon reaching desorption pressure, the heating phase starts and heat at $80\text{--}100^\circ\text{C}$ is supplied to the adsorption column by a heating jack using water as the heat transfer medium. The heating jack heats thermal masses and drives endothermic desorption of CO_2 and H_2O . The desorption phase starts when the sorbent has reached the desired desorption temperature. In the desorption phase, heat is continuously supplied to further drive endothermic desorption of CO_2 and H_2O , thus regenerating the sorbent. The exact temperature and pressure of desorption are determined via optimisation (see Sect. 3.3). In the subsequent cooling phase, water at ambient temperature is introduced into the heating jack. Thereby, the temperature in the adsorption column is cooled below 90°C before starting with the next adsorption phase. The cooling phase is necessary to avoid excessive oxidative degradation of the sorbent [29, 30].

The product stream exiting the adsorption column in the desorption phase consists of CO_2 and water vapour. In this study, the water content of the product stream is reduced by cooling the product stream and condensing the water vapour at ambient temperature and pressure. A dry cooler is used to dissipate the heat of the

product stream and is modeled as part of the DAC unit. An electrical heat pump supplies heat during the desorption phase. The electrical energy needed to drive the heat pump is calculated based on the ideal coefficient of performance (COP_{carnot}) and exergetic and mechanical efficiencies of the heat pump. The ideal coefficient of performance COP_{carnot} depends on the temperature of desorption and the ambient temperature.

For CO_2 storage, the electricity demand for compressing the desorbed CO_2 stream from ambient pressure (1 bar) to injection pressure (216 bar) is considered. The injection pressure is determined for an injection site with a depth of 2000 m [21, 22]. The electricity demand for compression is calculated based on an isothermal flow, assuming the desorbed CO_2 stream is an ideal gas. Since the DAC plant siting is flexible, we assume that the DAC plant is collocated with a storage site neglecting CO_2 transport. For a further assessment of DACCS siting, we include an assessment of CO_2 storage potential via CO_2 mineralisation. However, a detailed geological survey will be necessary for a reliable decision on the siting of a DACCS plant. Therefore, a detailed assessment of CO_2 storage and transport is out of the scope of our study, but our framework allows for easily integrating regionalised storage and transport data once available.

3.2 Assessment of DACCS performance

Three commonly used performance metrics to assess the thermodynamic and environmental performance of DACCS are SED [6, 10, 13, 15, 16, 22], productivity [6, 10, 13, 15, 16, 22], and CRE [18, 20, 21]. The performance metric SED measures the energy demand of a DACCS plant per gross mass of captured CO_2 in one cycle. The productivity of DACCS divides the gross mass of CO_2 captured in one cycle ($m_{CO_2, \text{captured}}$) by the cycle duration $\Delta\tau$:

$$\text{Productivity} = \frac{m_{CO_2, \text{captured}}}{\Delta\tau}.$$

The CRE relates the net mass of CO_2 captured in one cycle ($m_{CO_2, \text{captured}} - m_{CO_2-\text{eq,process}}$) to the gross mass of CO_2 captured in one cycle ($m_{CO_2, \text{captured}}$):

$$\text{CRE} = \frac{m_{CO_2, \text{captured}} - m_{CO_2-\text{eq,process}}}{m_{CO_2, \text{captured}}}.$$

Here, the net mass of captured CO_2 is calculated by subtracting process emissions of DACCS ($m_{CO_2-\text{eq,process}}$) from the gross mass of captured CO_2 .

Our recent study shows a trade-off between CRE and productivity [22]: Process designs with high CRE-values result in lower productivity and vice versa. Accounting for this trade-off is crucial for holistically assessing the potential of DACCS. Our recent study assesses this trade-off using a dynamic model combined with multi-objective optimisation. However, multi-objective optimisation has high computational costs, which is a barrier to a global analysis. To allow for a global analysis, we use the CRR as a single objective function in this study. The CRR is defined as the product of CRE and productivity, resulting in:

$$\text{CRR} = \frac{m_{CO_2, \text{captured}} - m_{CO_2-\text{eq,process}}}{\Delta\tau}. \quad (1)$$

The CRR can be interpreted as the rate at which DACCS generates net negative emissions when accounting for the life cycle GHG emissions of DACCS. Hence, it is a combined measure of thermodynamic and environmental performance.

3.3 Optimisation of DACCS performance

An optimisation problem (Eq. 2) is formulated with the CRR as the single objective function to ensure the optimal design of the DACCS process for various ambient conditions and electricity inputs. Optimal design refers to optimising free design parameters of the DACCS process in order to maximise the CRR. Here, the free design parameters are the duration of the adsorption, heating, and desorption phase, desorption temperature, desorption pressure, and air mass flow rate during adsorption. The free design parameters are grouped in the design vector p_d . We provide a list with the lower and upper bounds of all free design parameters in the supporting information (see Table S1).

$$\begin{aligned} & \max_{x(\cdot), z(\cdot), p_d} \quad \text{CRR}(x(\tau_{\text{cycle}}), z(\tau_{\text{cycle}}), T_{\text{amb}}, \varphi_{\text{amb}}, CF_{\text{el}}, p_d) \quad (\text{objective function}) \\ & \text{s.t.} \quad \dot{x} = f(x(\tau), z(\tau)) \quad (\text{dynamic model}) \\ & \quad \quad 0 = g(x(\tau), z(\tau)) \\ & \quad \quad x(\tau = 0) = x(\tau = \tau_{\text{cycle}}) \quad (\text{cyclic steady - state}) \end{aligned} \quad (2)$$

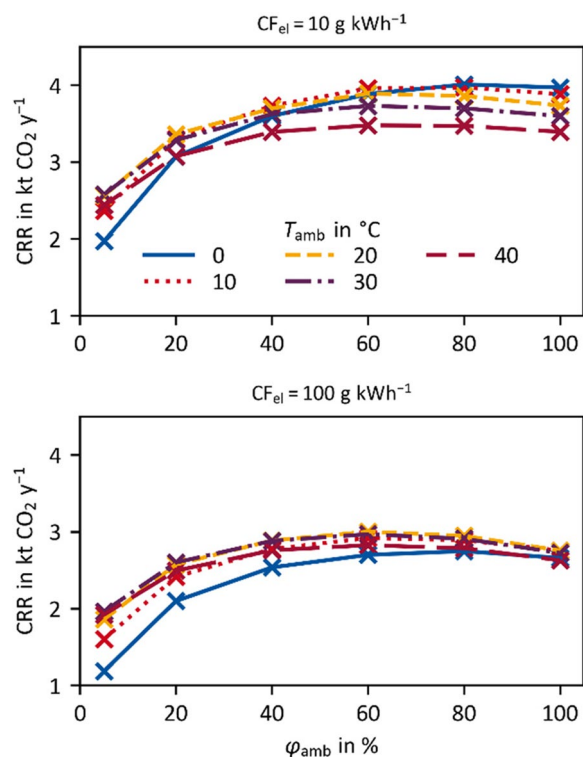


Fig. 3 Performance diagrams of the maximal carbon removal rate (CRR) of adsorption-based direct air carbon capture and storage (DACCS) for two carbon footprints of the electricity supply (CF_{el}): (top) $\text{CF}_{\text{el}} = 10 \text{ g kWh}^{-1}$ and (bottom) $\text{CF}_{\text{el}} = 100 \text{ g kWh}^{-1}$. The CRR is given as a function of relative humidity (φ_{amb}) for different ambient temperatures (T_{amb}). Results for extreme temperature values of -10 and 50°C are excluded to increase readability. Markers indicate optimised data points. Linear interpolation is used between the optimised data points

The optimisation problem is constrained by the system of equations describing the dynamic model of the DACCS system and a cyclic steady-state condition. The differential states of the dynamic model are summarised in x and the algebraic states in z . The model is optimised using a genetic optimisation algorithm included in the Python-based DEAP toolbox [31].

The optimisation problem is solved 378 times for different triples of ambient temperature, relative humidity, and electricity carbon footprint. The ambient temperature ranges from -10 to 50°C , relative humidity from 5 to 100% , and carbon footprint from 0 to $1300 \text{ g CO}_2 \text{ kWh}^{-1}$. Subsequently, we use linear interpolation to obtain CRR-values between the optimised data points. The ambient temperature, relative humidity, and carbon footprint of the electricity supply stay constant throughout each optimisation. Thus, we assume constant ambient conditions during each DAC cycle. We are aware that temperature and humidity likely change during one cycle since cycle durations are between 3 and 8 h , reaching up to 22 h in

extreme cases. However, accounting for diurnal variations in the optimisation significantly increases computational effort and is therefore out of the scope of the analysis.

Figure 3 shows the resulting performance diagrams for two carbon footprints of the electricity supply. Performance diagrams for all investigated carbon footprints and temperatures can be found in the supporting information (Figures S1 and S2).

In the range of $\varphi_{\text{amb}} = 5 - 60\%$, the CRR increases with increasing relative humidity for all ambient temperatures. Above $\varphi_{\text{amb}} = 60\% - 80\%$, the CRR decreases slightly with a further increase in relative humidity for all ambient temperatures except 0°C . These trends are caused by two opposing effects: On the one hand, CO_2 capacity is enhanced by the co-adsorption of water. Note that the co-adsorption behaviour is specific to the considered adsorbent material APDES-NFC and can be largely different for other materials. On the other hand, co-adsorbed water must be desorbed, increasing energy demand for regeneration. Both effects have been reported in the literature [9, 10, 13]. The larger CO_2 capacity reduces SED and increases productivity, thus reducing emissions from the DACCS process. Hence, the enhanced CO_2 capacity increases CRR. On the contrary, desorbing co-adsorbed water requires large amounts of energy, increasing SED and decreasing CRR.

The temperature has a dual effect on the CRR. On the one hand, lower temperatures enhance the equilibrium loading of the sorbent, which increases the CO_2 and H_2O capacity. On the other hand, lower ambient temperatures increase the temperature lift of the heat pump, resulting in a lower COP and CRR. The decrease in COP with decreasing temperature has smaller impacts on the CRR for small carbon footprints. Hence, lower ambient temperatures positively affect the CRR for small carbon footprints.

Furthermore, lower temperatures increase productivity and thus CRR, which can be attributed to utilising a first-order linear driving force (LDF) kinetic model with a temperature-independent mass transfer coefficient. In the LDF model, adsorption kinetics are solely determined by the mass transport of the sorptive: Mass transport is driven by the difference between the sorbent's equilibrium loading and current loading. Hence, lower ambient temperatures accelerate adsorption as they increase equilibrium loading. The LDF model has been employed in recent numerical studies on DAC [10, 13, 22], and it was demonstrated that the LDF model fits experimental data very well for the investigated sorbent [13]. Hence, we use the LDF in our analysis.

For higher electricity carbon footprints, the general trend of the performance diagram stays the same (Figures S1 and S2). However, CRR decreases with an increasing carbon footprint of electricity.

3.4 Application of the performance diagrams to assess global DACCS performance

To determine the regional CRR, we use global hourly weather data from 2015 to 2019 and regional data on the carbon footprint of electricity for the four investigated supply scenarios. In addition, we investigate the effect of temporal weather data resolution on the CRR (Section S8). The sensitivity study shows that using monthly and yearly averaged weather data overestimates the CRR. Both ambient temperature and humidity are subject to day-night variations, which are smoothed out with monthly or yearly averaging. The performance diagrams (Fig. 3) show that ambient temperature and humidity nonlinearly affect the CRR. This nonlinearity is not represented when using averaged data. Hence, we use hourly weather data in our analysis.

Global hourly weather data was taken from NASA's MERRA-2 [32] dataset. The dataset provides data with a spatial resolution of $0.5^\circ \text{ lat} \times 0.625^\circ \text{ lon}$. Linear interpolation is used to calculate the CRR for triples of ambient temperature, relative humidity, and electricity carbon footprint within the limits of the optimised data points. The CRR of every hour of the investigated years is calculated based on the closest optimised data point for triples outside the limits. For example, the CRR is set to the CRR value at -10° C for ambient temperatures below -10° C . For the final global maps, we average the hourly-resolved CRR over the investigated years.

4 Modelling life cycle emissions of the electricity supply

The carbon footprint of electricity varies strongly depending on source of electricity and region. In this work, four electricity supply scenarios are exemplarily investigated (see Sect. 2): (a) Ideal electricity supply, (b) grid, (c) solar PV, and (d) wind electricity. All data on carbon footprints is taken from the LCA database ecoinvent version 3.9.1 [23] using the cut-off system model and the life cycle assessment method Environmental Footprint 3.1. The used ecoinvent datasets are given in Table S2 of the supporting information.

For the ideal electricity supply scenario, the carbon footprint of electricity is 0 g kWh^{-1} . Thus, we isolate the effect of regional climate on DACCS performance. Ecoinvent provides data on the carbon footprint of grid electricity for 136 countries. Grid electricity includes various electricity sources, such as coal, natural gas, hydroelectric, wind, solar, and nuclear power. The ecoinvent data

set represents the carbon footprint of grid electricity in 2014–2022. Carbon footprint data is resolved on multiple electricity sub-grids for Brazil, Canada, China, India, and the USA [23]. However, geospatial information on sub-grids is lacking; hence, sub-grid carbon footprints cannot be matched to specific coordinates. Therefore, we average the carbon footprint over all sub-grids and assume a uniform carbon footprint for the respective country.

For PV and wind power plants, we first determine GHG emissions from construction and end-of-life based on ecoinvent. We choose one proxy power plant for PV and wind electricity generation as the basis of our investigations. For the PV power plant, we select a freestanding 570 kWp plant without sun tracking. For the wind power plant, we use an Enercon E-82/2300 with a peak power output of 2.3 MW and a hub height of 108 m. We use these power plants as proxies as ecoinvent provides comprehensive LCA datasets for both. We are aware that the selected wind power plant is not optimal for all regions. Selecting different wind power plants depending on the location could increase the accuracy of the calculated wind energy yields. Currently, LCA data on different types of wind turbines is very limited. Therefore, using different turbines would introduce larger errors in the LCA data sets. For this reason, we use the specified wind power plant as the proxy.

Using these proxy power plants, we calculate the average electricity output from 2015 to 2019. To determine the regionalised energy output of the PV power plant, we use PVGIS [33], provided by the European Union's Joint Energy Research Center. To determine the regionalised energy output of the wind power plant, we use a similar approach to that of PV. We calculate the average wind energy yield of the selected wind power plant based on global wind speed data from 2015 to 2019. For this purpose, we use the Python library windpowerlib [34]. Windpowerlib provides power curves for the investigated wind power plant, correlating wind speed and wind power output. Section S5 of the supporting information thoroughly describes the modeling of PV and wind electricity carbon footprints.

Using the approach described above, we derive global maps of the carbon footprint of grid, PV, and wind electricity (Fig. 4). The lowest carbon footprints are found in the order of wind electricity, PV electricity, and grid electricity.

The carbon footprint of grid electricity ranges from below $40 \text{ g CO}_2 \text{ kWh}^{-1}$ for countries with large shares of renewable energy, like Paraguay, Norway, and Switzerland, to above $900 \text{ g CO}_2 \text{ kWh}^{-1}$ for countries heavily relying on fossil-based power plants. Greyed-out

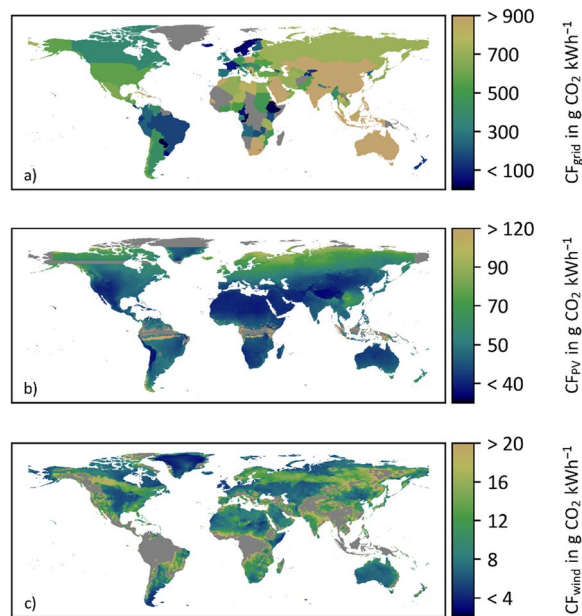


Fig. 4 Global maps of the carbon footprints (CF) of (a) grid, (b) PV, and (c) wind electricity. In sub-figures (a) and (b), greyed-out areas indicate that no data is available. In sub-figure (c), greyed-out areas are excluded from the analysis due to inefficient wind turbine uses. Note that the limits of the colorbars differ as carbon footprints vary widely for the investigated electricity sources

countries indicate no ecoinvent data on the carbon footprint of grid electricity.

The carbon footprint of PV electricity ranges from 41 to above 120 g CO₂ kWh⁻¹, with greyed-out areas indicating a lack of PVGIS data. As expected, the lowest carbon footprints are found in sunny regions, such as North Africa, the Middle East, and the southwest of the USA. High carbon footprints are mostly located in northern regions due to lower solar irradiation.

The carbon footprint of wind electricity ranges from 3 to above 20 g CO₂ kWh⁻¹. The lowest values are found in Greenland, Iceland, and the southern tip of South America. The highest carbon footprints are located close to forests, such as the boreal forests in North America, Europe, and Asia, as well as the African rainforest. Greyed-out regions are excluded from the analysis since the investigated wind turbine achieved a capacity factor of less than 15%. For comparison, the International Renewable Energy Agency (IRENA) stated that wind power plants had an average capacity factor of 34% globally in 2018 [35]. Hence, using the investigated wind power plant in excluded regions would lead to high carbon footprints and, most likely, uneconomical performance. Overall, the trend of wind carbon footprints agrees with data on mean wind power density from the global wind atlas [36]. Moreover, wind electricity carbon footprints of 3 to

20 g CO₂ kWh⁻¹ are mostly in line with datasets from the LCA databases ecoinvent and Gabi: In both databases, carbon footprints of wind electricity are in the range of 10 to 35 g CO₂ kWh⁻¹ depending on the type of power plant and country [23, 37].

5 Results and discussion

5.1 Scenario (a): ideal electricity supply scenario

Figure 5 shows the CRR for a DACCS plant powered by an ideal electricity supply (CRR_{ideal}) with a carbon footprint of 0 g CO₂ kWh⁻¹.

Figure 5 shows that the highest CRR_{ideal}-values are mostly found in the north. A comparison of Fig. 5 to the global maps of average temperature and humidity (Figures S6 and S7) reveals that highest CRR_{ideal}-values are found in regions with low ambient temperature and high relative humidity. This observation is consistent with the performance diagram (Fig. 3), as cold temperatures and high relative humidity values increase the CRR_{ideal} of DACCS. On the other hand, in hot and dry regions DACCS exhibits a very low CRR_{ideal}. For example, in the North African interior and Australia very low CRR_{ideal} values are found in dry desert regions with an average ambient temperature of 30 °C and average relative humidity of 20–30% (Figures S6 and S7). For the investigated DACCS plant, CRR_{ideal} varies from 3.17 kt CO₂ y⁻¹ at one site in Chad to 4.17 kt CO₂ y⁻¹ at one site in Greenland. Hence, optimal DACCS placement can increase CRR_{ideal} by more than 32% when comparing the best and worst locations. This improvement can be entirely attributed to varying regional ambient conditions, as the carbon footprint of electricity stays constant in this analysis. The increase in CRR_{ideal} shows the importance of accounting for the impact of regional climate on DACCS performance.

To confirm the findings, we compare our results to a global assessment of the SED and productivity of DAC

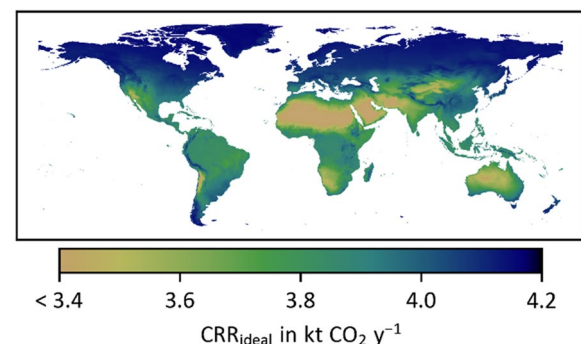


Fig. 5 Carbon removal rate of the investigated DACCS plant using an ideal electricity supply (CRR_{ideal}) with a carbon footprint of 0 g CO₂ kWh⁻¹

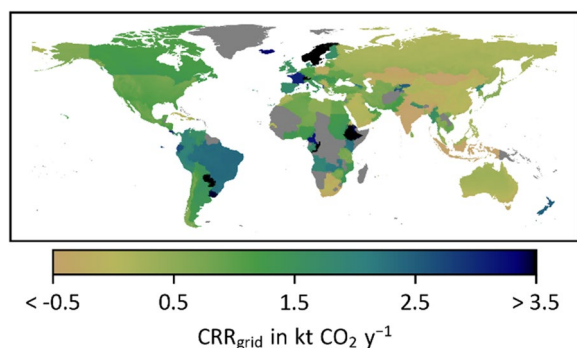


Fig. 6 Carbon removal rate of the investigated DACCS plant using grid electricity (CRR_{grid}). Greyed-out areas indicate no data availability

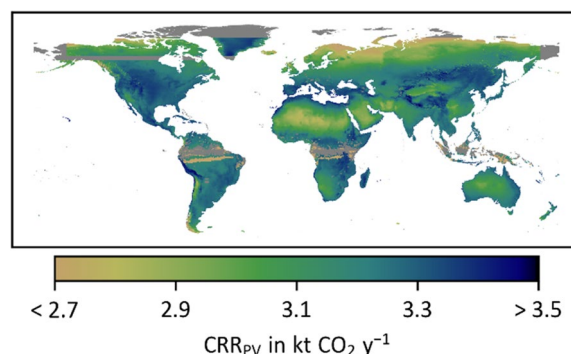


Fig. 7 Carbon removal rate of the investigated DACCS plant using PV electricity (CRR_{PV}). For greyed-out areas, no data is available through PVGIS

conducted by Sendi et al. [16]. For this purpose, we optimise our model using SED and productivity as objective functions (Figures S8 and S9). The maps provided by Sendi et al. and our results largely agree when compared visually, further supporting our findings.

5.2 Scenario (b): grid electricity supply scenario

Figure 6 shows the CRR of DACCS powered by grid electricity (CRR_{grid}).

The CRR_{grid} varies widely from -1.40 to $3.73 \text{ kt CO}_2 \text{ y}^{-1}$. The CRR_{grid} is positive in most countries, highlighting that negative emissions can be achieved using grid electricity in 136 countries. However, the CRR_{grid} is mostly small, making DACCS relatively inefficient. Countries such as Norway, Sweden, and Iceland are an exception, with DACCS already reaching up to 89% of the maximal achievable CRR_{ideal} . In these countries, the carbon footprint of grid electricity hardly limits the CRR, even with grid electricity. On the other hand, the CRR_{grid} is negative in 20 countries due to large shares of fossil electricity generation. Hence, DACCS would lead to additional emissions instead of negative ones.

A comparison of the CRR_{grid} and the carbon footprint of grid electricity (Fig. 4) suggests that CRR_{grid} is mainly affected by the carbon footprint of the electricity supply. For example, cold regions with high relative humidity in Russia, and the U.S., almost optimal for DACCS using carbon free electricity (Fig. 5), exhibit very low CRR_{grid} -values due to the high carbon intensity of grid electricity. In contrast, the impact of regional climate on DACCS performance has little influence on the CRR_{grid} . Therefore, countries with the highest CRR_{grid} coincide with countries exhibiting low electricity carbon footprints.

5.3 Scenario (c): PV electricity supply scenario

Figure 7 shows the CRR of DACCS powered by PV electricity (CRR_{PV}). The CRR_{PV} ranges from $2.16 \text{ kt CO}_2 \text{ y}^{-1}$

at one site in Pakistan to $3.53 \text{ kt CO}_2 \text{ y}^{-1}$ at one site in the USA. Thus, negative emissions are achieved for all global regions when PV electricity is used to drive DACCS. Still, accounting for regional climate can increase CRR_{PV} by up to 63% when comparing the best and worst DACCS sites.

The highest CRR_{PV} -values are found northwest of the USA, Greenland, and the North African coast. In contrast, the lowest CRR_{PV} -values are found in the North African interior and the northern parts of Europe and Asia. The northwest of the USA has relatively low carbon footprints of PV electricity (40 to $50 \text{ g CO}_2 \text{ kWh}^{-1}$, Fig. 4) and relatively high CRR_{ideal} -values of DACCS (Fig. 5). However, the region is neither optimal for PV power plants nor DACCS. Still, the northwest of the USA is one of the best regions for PV-powered DACCS with a CRR_{PV} of approximately 3.3 to $3.4 \text{ kt CO}_2 \text{ y}^{-1}$. Therefore, it is crucial to assess both DACCS and PV performance when selecting optimal sites for the operation of PV-powered DACCS.

Greenland has a relatively high carbon footprint of PV electricity, mostly 50 to $80 \text{ g CO}_2 \text{ kWh}^{-1}$ (Fig. 4). However, due to cold temperatures and high relative humidity, Greenland exhibits among the highest CRR_{ideal} -values (Fig. 5). In Greenland, the enhanced performance of DACCS under cold and humid climates outweighs the relatively high carbon footprint of PV electricity, which leads to very high CRR_{PV} -values.

North Africa generally shows very low carbon footprints of PV electricity (Fig. 4) due to high solar irradiation. Additionally, at the North African coast, the average relative humidity is in the range of 40 to 60% , which results in relatively good DACCS performance. Accordingly, low CRR_{PV} -values could be observed on the North African coast. On the other hand, the North African interior shows among the lowest CRR_{PV} -values despite the low carbon footprint of PV electricity: This is because the Sahara, one of the hottest and

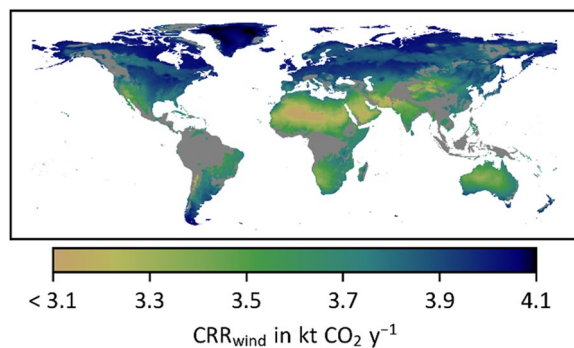


Fig. 8 Carbon removal rate of the investigated DACCS plant using wind electricity (CRR_{wind}). Greyed-out countries are excluded from the assessment

driest regions in the world, is located in the North African interior. As hot and dry climates reduce the CRR (Fig. 3), CRR_{PV} is lowered significantly. Hence, the climate impact on DACCS performance outweighs the low carbon footprint of PV electricity in the Sahara.

5.4 Scenario (d): wind electricity supply scenario

Figure 8 shows the CRR of DACCS powered by wind electricity (CRR_{wind}). When comparing the worst and best locations, the CRR_{wind} increases by 39% from 2.95 kt CO₂ y⁻¹ at one site in Chile to 4.1 kt CO₂ y⁻¹ at one site in Greenland. Hence, like PV-powered DACCS, wind-powered DACCS can achieve negative emissions in all investigated locations.

Optimal regions for wind-powered DACCS operation are mostly collocated with optimal regions found in the assessment of the CRR_{ideal} . The carbon footprint of wind electricity is much smaller than that of grid or PV electricity for all investigated locations (see Fig. 4). Thus, the share of emissions from electricity generation on overall life cycle emissions of the DACCS plant is also much smaller for wind-powered DACCS. Hence, regional differences in the carbon footprint of wind electricity only have a small impact on the CRR. Therefore, in the context of a global analysis, the maps for CRR_{wind} and CRR_{ideal} are very similar.

Nevertheless, there are regions where favourable ambient conditions for DACCS and wind power plants overlap, resulting in the highest CRR_{wind} -values. The highest CRR_{wind} -values are in Iceland and Greenland, which exhibit low ambient temperatures, high relative humidity, and high wind speeds, making them ideal for DACCS and wind power plants. The lowest CRR_{wind} -values are again found in, e.g., the North African interior, the Middle East, and Australia due to the negative impact of hot and dry climates on DAC performance.

5.5 CO₂ storage

Ideally, DAC plants should be located close to potential CO₂ storage locations as CO₂ transport can significantly reduce the climate benefit of DACCS [21]. Typically, CO₂ is stored via injection in a gaseous, liquid or supercritical state into underground reservoirs [38]. CO₂ trapping can occur in multiple ways [38, 39]: For example, it can be trapped beneath an impermeable cap rock, in small rock pores, or react with underground minerals to form solid carbonates. CO₂ storage via mineral carbonation is especially promising as the CO₂ is stored in solid form, significantly reducing the risk of leakage [38, 39].

The company Carbfix developed a method to store CO₂ as solid carbonate by significantly enhancing the natural mineralisation process. First, the CO₂ is dissolved in water, and then the solution is injected into suitable reactive rocks that promote CO₂ mineralisation [38]. The potential for CO₂ storage via mineralisation varies greatly based on regional geology. Carbfix provides a global map showing favourable regions for storing CO₂ via CO₂ mineralisation [24]. The map is accessible through the Environmental Systems Research Institute [25]. In the supporting information, we show the onshore regions of that map (Figure S14). We want to emphasise that this map is a first indication of the feasibility of CO₂ storage via mineralisation, neglecting important factors such as water availability or bedrock permeability.

Figure 9 shows the intersection of Carbfix's map on CO₂ storage potential with our global DACCS assessment for the four investigated electricity supply scenarios (see Sects. 5.1 – 5.4).

When using an ideal electricity supply, the highest CRR_{ideal} -values are mostly found in cold climates in the northern hemisphere. Many of these regions intersect with potential sites for CO₂ storage via CO₂ mineralisation (Fig. 9a). For example, Canada, Scandinavia, Iceland, and Russia have large areas suited for both DAC and CO₂ storage.

Optimal sites for DACCS driven by grid electricity and potential CO₂ storage intersect, e.g., in Scandinavia, Brazil, and Ethiopia (Fig. 9b). However, most of the storage potential is located in countries with GHG-intensive electricity grids, thus limiting the climate benefit of DACCS. Sites suited for PV-driven DACCS and potential CO₂ storage can be found in the northwest of the USA, Ethiopia and Kenya. Yet, again the largest areas with storage potential are located in regions less favourable for PV-driven DACCS, such as Russia, Alaska, and India. Cold regions in the northern hemisphere, such as Iceland, Greenland, and parts of Russia, show the highest potential for wind-powered DACCS. Figure 9d shows

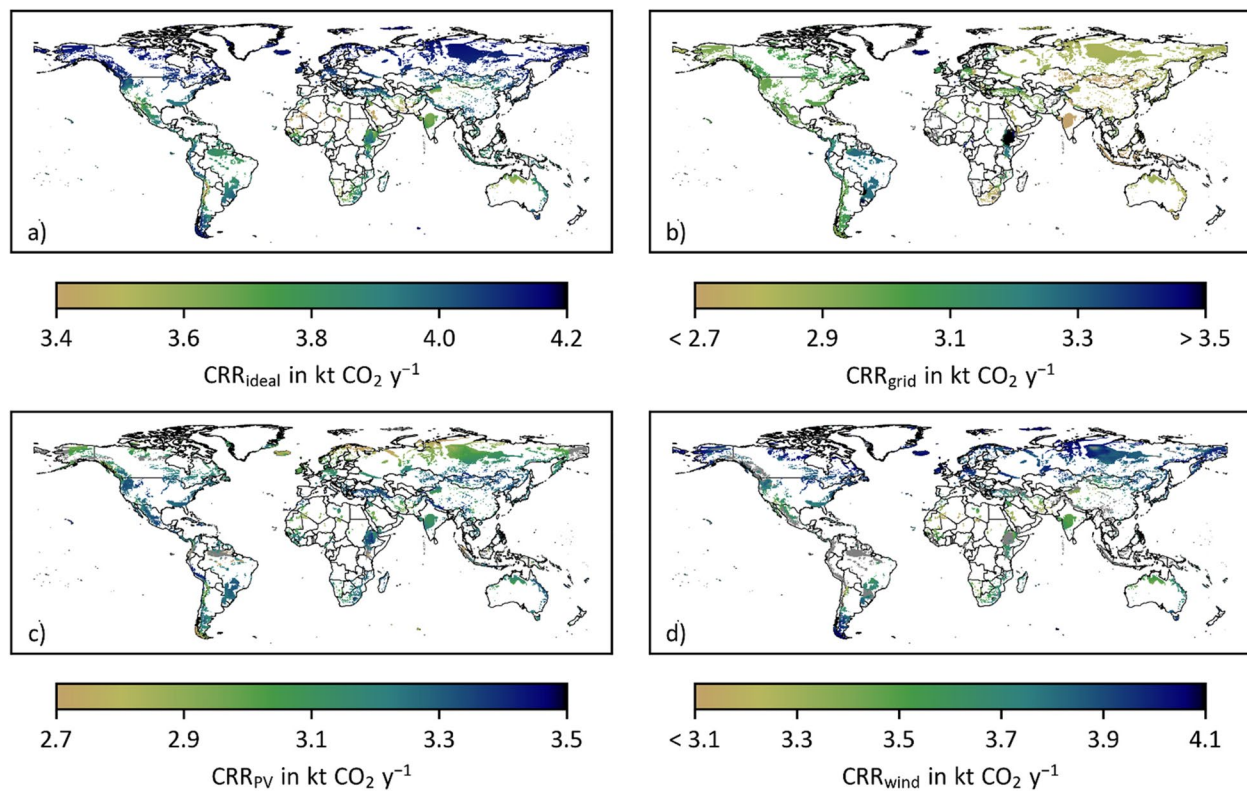


Fig. 9 Assessment of the CRR of DACCS showing only onshore regions with a favourable geology for CO₂ storage via CO₂ mineralisation. The figures show the intersection of the global maps of the CRR of DACCS developed in the present study (see Sects. 5.1 – 5.4) with a global map of regions favourable for CO₂ mineralisation. The map on favourable geology for CO₂ mineralisation was published by Carbfix via the Environmental Systems Research Institute [25]

that Iceland and Russia especially exhibit large areas potentially suited for CO₂ storage, while Greenland's CO₂ storage potential is very limited.

The analysis shows that regional CO₂ storage potential strongly influences the optimal site selection for DACCS. Hence, it is essential to include the potential for CO₂ storage to make a comprehensive siting decision.

5.6 Limitations and outlook

Our study has two main limitations:

- (1) Equilibrium and kinetic data on sorbents suitable for DAC is very limited, especially lacking data on the co-adsorption of CO₂ and H₂O. To showcase our method, we use an empirical model derived by Stampi-Bombelli et al. for the sorbent APDES-NFC. Therefore, the quantitative results of our study are only valid for this specific sorbent. The regional performance of DAC using other materials may vary considerably.
- (2) We conduct a limited assessment of regional CO₂ storage potential due to a lack of regionalised CO₂

storage data. For this reason, we only include CO₂ mineralisation as a potential CO₂ storage technology to showcase our method. Moreover, the data on CO₂ mineralisation does not account for water availability and bed rock permeability, which is essential for a detailed assessment of CO₂ storage potential.

Due to the flexibility of our method, these limitations can be addressed in the future. We recommend that our method is applied to different DAC-relevant sorbents, which requires further research on characterising the co-adsorption of CO₂ and H₂O for different sorbents. Also, the linear scaling of the DAC model from laboratory to kt-scale should be improved by an accurate model of a real DAC plant.

To further improve the assessment of DACCS, more detailed data on CO₂ storage should be considered. Especially, other methods to store CO₂, such as CO₂ trapping in porous rocks or beneath an impermeable cap rock, should be included in the analysis. With the addition of more detailed CO₂ storage data, different modes and

routes of CO₂ transport can be included in the analysis to improve site selection for DACCS plants.

Additionally, daily fluctuations in ambient conditions and renewable energy supply should be accounted for in future analyses [40]. On a global scale, such analyses demand a prohibitive amount of computational resources and likely require a simplified DACCS model.

Finally, we have demonstrated a method for an initial assessment of the climate benefit of DACCS based on the CRR as a thermo-ecological criterion. Still, for a detailed siting decision of a DACCS plant, important real-world conditions have to be considered, e.g. infrastructure, social acceptance, and economic feasibility. Our techno-ecological analysis does not allow a conclusion on the first two aspects, but may provide a first indication of the regional economic feasibility of DACCS.

It is to be expected that net costs of CO₂ removal will also vary by region. For example, a higher CRR leads to lower costs when the same economic conditions are assumed. More specifically, a high CRR means that the net amount of CO₂ removed per unit of time is increased. Given that the net amount of CO₂ removed can be sold via carbon removal credits, a high CRR may decrease the amortisation time of the DACCS plant. The lower amortisation time can directly result in a higher economic feasibility. We want to highlight that the method presented here is flexible in that an economic objective function can be used. Therefore, we recommend extending the analysis by a techno-economic assessment of DACCS.

6 Conclusion

We present a comprehensive method for an integrated global analysis of adsorption-based direct air carbon capture and storage (DACCS) plants. The method considers climate-dependent direct air capture (DAC) performance, regional carbon footprints of the electricity supply, and regional CO₂ storage potential for CO₂ mineralisation. As performance metric, we use the carbon removal rate (CRR), representing the net mass of CO₂ removed from the atmosphere per unit of time when accounting for the life cycle GHG emissions of the DACCS plant. As case study, we investigate four electricity supply scenarios. The scenarios comprise a carbon-free ideal electricity supply as well as grid, PV, and wind electricity.

Optimal siting of DACCS by leveraging the impact of regional climate on DAC performance can increase the CRR_{ideal} from 3.17 to 4.17 kt CO₂ y⁻¹ (32%) when using a carbon-free source of electricity. Grid-powered DACCS achieves positive CRR_{grid}-values in most countries, already enabling negative emissions using grid electricity in 136 countries. However, the carbon footprint of grid electricity is often high due to large shares of fossil energy, making DACCS quite inefficient and slow regarding CRR.

DACCS powered by PV and wind electricity achieves high CRR-values in all investigated regions. Optimal siting can increase the CRR_{PV} by 63% from 2.16 to 3.53 kt CO₂ y⁻¹ and the CRR_{wind} by 39% from 2.95 to 4.1 kt CO₂ y⁻¹ when comparing worst and best locations. For DACCS powered by PV electricity, we find a complex competition between regional DAC and PV performance. Often, optimal conditions for the operation of DAC and PV are opposed. For wind-powered DACCS, the climate impact on DAC performance dominates the CRR_{wind}, as wind energy is generally a very clean energy source in all investigated regions. Moreover, sites with high wind speeds even overlap with optimal locations for DAC performance.

Finally, we extend our analysis of DACCS by accounting for regional CO₂ storage potential through CO₂ mineralisation. Our analysis shows that the number of optimal sites for DACCS is significantly reduced if CO₂ storage is to be co-located. Therefore, it is essential to include the potential for CO₂ storage when siting DACCS plants.

Overall, our analysis shows that optimal siting of DACCS can significantly improve its carbon removal potential and thus climate benefit.

Nomenclature

$m_{\text{CO}_2, \text{captured}}$	Mass of CO ₂ captured by the DAC process in kg
$m_{\text{CO}_2 - \text{eq, process}}$	Mass of CO ₂ emitted over the life cycle of the DACCS process in kg
φ_{amb}	Ambient relative humidity in %
$\dot{Q}$	Heat flow rate in W
T_{amb}	Ambient temperature in °C
$\dot{W}_{\text{el}}$	Electric power in W

Abbreviations

GHG	Greenhouse gas
DAC	Direct air capture
DACCS	Direct air carbon capture and storage
SED	Specific energy demand
CRE	Carbon removal efficiency
LCA	Life cycle assessment
CRR	Carbon removal rate
CF	Carbon footprint

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s43979-024-00100-z>.

Supplementary Material 1.

Acknowledgements

Not applicable.

Authors' contributions

DR: conceptualization, data curation, formal analysis, investigation, software methodology, visualization, writing – original draft, writing – review & editing. PP: conceptualization, software, formal analysis, investigation, methodology. ME: supervision, conceptualization, data curation, investigation, methodology, writing – review & editing. LM: conceptualization, data curation, writing – review & editing. NvdA: supervision, conceptualization, methodology, visualization, writing – review & editing. All authors read and approved the final manuscript.

Funding

This work has been carried out within the project "DAC-TALES—Transdisciplinary Assessment Combining Labs, the Environment, the Economy, and Society" (01LS2106A). The project is funded by the German Federal Ministry of Education and Research (BMBF).

Availability of data and materials

A detailed description of the energy supply modelling can be found in the supporting information, which also contains all CRR performance diagrams used for the global assessment of DACCS. The MERRA dataset is available at https://cmr.earthdata.nasa.gov/search/concepts/C1276812863-GES_DISC.html. PV-Gis is available at https://re.jrc.ec.europa.eu/pvg_tools/en/. The data on regional CO₂ storage potential is available at [7].

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

There are no conflicts of interest to declare.

Received: 16 April 2024 Revised: 24 July 2024 Accepted: 19 August 2024
Published online: 09 September 2024

References

- Intergovernmental Panel on Climate Change (2018) Global warming of 1.5°C - An IPCC Special Report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty
- Buck HJ, Carton W, Lund JF, Markusson N (2023) Why residual emissions matter right now. *Nature Clim Change* 13:351–358. <https://doi.org/10.1038/s41558-022-01592-2>
- Fawzy S, Osman AI, Doran J, Rooney DW (2020) Strategies for mitigation of climate change: a review. *Environ Chem Lett* 18:2069–2094. <https://doi.org/10.1007/s10311-020-01059-w>
- McQueen N, Gomes KV, McCormick C, Blumanthal K, Pisciotto M, Wilcox J (2021) A review of direct air capture (DAC): scaling up commercial technologies and innovating for the future. *Prog Energy* 3:32001. <https://doi.org/10.1088/2516-1083/abf1ce>
- Bouaboula H, Chaouki J, Belmabkhout Y, Zabout A (2024) Comparative review of Direct air capture technologies: From technical, commercial, economic, and environmental aspects. *Chem Eng J* 484:149411. <https://doi.org/10.1016/j.cej.2024.149411>
- Fasihi M, Efimova O, Breyer C (2019) Techno-economic assessment of CO₂ direct air capture plants. *J Clean Prod* 224:957–980. <https://doi.org/10.1016/j.jclepro.2019.03.086>
- Schlosser F, Jesper M, Vogelsang J, Walmsley TG, Arpagaus C, Hesselbach J (2020) Large-scale heat pumps: applications, performance, economic feasibility and industrial integration. *Renew Sustain Energy Rev* 133:110219. <https://doi.org/10.1016/j.rser.2020.110219>
- Ješić D, Lašić Jurković D, Pohar A, Suhadolnik L, Likozar B (2021) Engineering photocatalytic and photoelectrocatalytic CO₂ reduction reactions: Mechanisms, intrinsic kinetics, mass transfer resistances, reactors and multi-scale modelling simulations. *Chem Eng J* 407:126799. <https://doi.org/10.1016/j.cej.2020.126799>
- Wurzbacher JA, Gebald C, Piatkowski N, Steinfeld A (2012) Concurrent separation of CO₂ and H₂O from air by a temperature-vacuum swing adsorption/desorption cycle. *Environ Sci Technol* 46:9191–9198. <https://doi.org/10.1021/es301953k>
- Young J, García-Díez E, García S, van der Spek M (2021) The impact of binary water–CO₂ isotherm models on the optimal performance of sorbent-based direct air capture processes. *Energy Environ Sci* 14:5377–5394. <https://doi.org/10.1039/d1ee01272j>
- Kovačič Ž, Likozar B, Huš M (2020) Photocatalytic CO₂ reduction: a review of Ab Initio Mechanism, Kinetics, and Multiscale Modeling Simulations. *ACS Catal* 10:14984–15007. <https://doi.org/10.1021/acscatal.0c02557>
- Pavličić A, Huš M, Prašnikar A, Likozar B (2020) Multiscale modelling of CO₂ reduction to methanol over industrial Cu/ZnO/Al₂O₃ heterogeneous catalyst: Linking ab initio surface reaction kinetics with reactor fluid dynamics. *J Clean Prod* 275:122958. <https://doi.org/10.1016/j.jclepro.2020.122958>
- Stampi-Bombelli V, van der Spek M, Mazzotti M (2020) Analysis of direct capture of CO₂ from ambient air via steam-assisted temperature–vacuum swing adsorption. *Adsorption* 26:1183–1197. <https://doi.org/10.1007/s10450-020-00249-w>
- Balasubramaniam BM, Thierry P-T, Lethier S, Pugnet V, Llewellyn P, Rajendran A (2024) Process-performance of solid sorbents for Direct Air Capture (DAC) of CO₂ in optimized temperature–vacuum swing adsorption (TVSA) cycles. *Chem Eng J* 485:149568. <https://doi.org/10.1016/j.cej.2024.149568>
- Schellevis M (2023) CO₂ capture from air : a process engineering approach. <https://doi.org/10.3990/1.9789036555852>
- Sendi M, Bui M, Mac Dowell N, Fennell P (2022) Geospatial analysis of regional climate impacts to accelerate cost-efficient direct air capture deployment. *One Earth* 5:1153–1164. <https://doi.org/10.1016/j.oneear.2022.09.003>
- Wiegner JF, Grimm A, Weimann L, Gazzani M (2022) Optimal design and operation of solid sorbent direct air capture processes at varying ambient conditions. *Ind Eng Chem Res* 61:12649–12667. <https://doi.org/10.1021/acs.iecr.2c00681>
- Deutz S, Bardow A (2021) Life-cycle assessment of an industrial direct air capture process based on temperature–vacuum swing adsorption. *Nat Energy* 6:203–213. <https://doi.org/10.1038/s41560-020-00771-9>
- de Jonge MM, Daemen J, Loriaux JM, Steinmann ZJ, Huijbregts MA (2019) Life cycle carbon efficiency of Direct Air Capture systems with strong hydroxide sorbents. *Int J Greenhouse Gas Control* 80:25–31. <https://doi.org/10.1016/j.jggc.2018.11.011>
- Madhu K, Pauliuk S, Dhathri S, Creutzig F (2021) Understanding environmental trade-offs and resource demand of direct air capture technologies through comparative life-cycle assessment. *Nat Energy* 6:1035–1044. <https://doi.org/10.1038/s41560-021-00922-6>
- Terlouw T, Treyer K, Bauer C, Mazzotti M (2021) Life cycle assessment of direct air carbon capture and storage with low-carbon energy sources. *Environ Sci Technol*. <https://doi.org/10.1021/acs.est.1c03263>
- Postweiler P, Engelpracht M, Rezo D, Gibelhaus A, von der Aßen N (2024) Environmental process optimisation of an adsorption-based direct air carbon capture and storage system. *Energy Environ. Sci* 17:3004–20
- Wernet G, Bauer C, Steubing B, Reinhard J, Moreno-Ruiz E, Weidema B (2016) The ecoinvent database version 3 (part I): overview and methodology. *Int J Life Cycle Assess* 21:1218–1230. <https://doi.org/10.1007/s11367-016-1087-8>
- Carbfix (2024) Mineral Storage Atlas. <https://www.carbfix.com/mineral-storage-atlas>. Accessed 12.06.24
- Esri (2024) Mineral Storage Atlas. <https://orkuveitan.maps.arcgis.com/apps/View/index.html?appid=ab4d5798f716486c9b9041277e97cf1d>. Accessed 12 June 2024
- Bau U, Lanzerath F, Gräber M, Graf S, Schreiber H, Thielen N, Bardow A (2014) Adsorption energy systems library - Modeling adsorption based chillers, heat pumps, thermal storages and desiccant systems. 10th International Modelica Conference:875–883. <https://doi.org/10.3384/ECP14096875>
- Institute of Technical Thermodynamics, RWTH Aachen University (2023) SorpLib - Adsorption Energy Systems Library. https://git.rwth-aachen.de/lt/SorpLib/-/tree/SorpLib_v2?ref_type=heads
- Sabatino F, Grimm A, Gallucci F, van Sint AM, Kramer GJ, Gazzani M (2021) A comparative energy and costs assessment and optimization for direct air capture technologies. *Joule* 5:2047–2076. <https://doi.org/10.1016/j.joule.2021.05.023>
- Fan Y, Jia X (2022) Progress in amine-functionalized Silica for CO₂ capture: important roles of support and amine structure. *Energy Fuels* 36:1252–1270. <https://doi.org/10.1021/acs.energyfuels.1c03788>

30. Jahandar Lashaki M, Khiavi S, Sayari A (2019) Stability of amine-function-alized CO₂ adsorbents: a multifaceted puzzle. *Chem Soc Rev* 48:3320–3405. <https://doi.org/10.1039/C8CS00877A>
31. Fortin FA, De Rainville FM, Gardner MA, Parizeau M, Gagné C (2012) Deap: Evolutionary algorithms made easy. *J Mach Learn Res* 13:2171–5
32. NASA Global Modeling and Assimilation Office (2022) MERRA-2 tavg1_2d_slv_Nx: 2d,1-Hourly,Time-Averaged,Single-Level,Assimilation,Single-Level Diagnostics V5.12.4
33. Huld T, Müller R, Gambardella A (2012) A new solar radiation database for estimating PV performance in Europe and Africa. *Sol Energy* 86:1803–1815. <https://doi.org/10.1016/j.solener.2012.03.006>
34. Haas S, Krien U, Schachler B, Shivam K, Bosch S (2021) windpowerlib - a python library to model wind power plants. <https://doi.org/10.5281/zenodo.4591809>. Accessed 22 Aug 2024.
35. International Renewable Energy Agency (2019) Future of wind: Deployment, investment, technology, grid integration and socio-economic aspects (A Global Energy Transformation paper)
36. Technical University of Denmark (2022) Global wind atlas 3.0. <https://globalwindatlas.info/>. Accessed Nov. 2022
37. thinkstep AG (2019) GaBi 9.2.0.58: Software-System and Database for Life Cycle Engineering, DB 8.7 - SP 39
38. Snæbjörnsdóttir SÓ, Sigfússon B, Marieni C, Goldberg D, Gislason SR, Oelkers EH (2020) Carbon dioxide storage through mineral carbonation. *Nat Rev Earth Environ* 1:90–102. <https://doi.org/10.1038/s43017-019-0011-8>
39. Kim K, Kim D, Na Y, Song Y, Wang J (2023) A review of carbon mineralization mechanism during geological CO₂ storage. *Heliyon* 9:e23135. <https://doi.org/10.1016/j.heliyon.2023.e23135>
40. Postweiler P, Engelpracht M, Rezo D, Nilge B, von der Aßen N (2023) Demand-side management for direct air carbon capture and storage: an enabler for low-cost negative emissions? Zenodo. <https://doi.org/10.5281/zenodo.10245218>

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.